

ECA0303 SIGNALS AND SYSTEM

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2. SIGNAL PROPERTIES AND TRANSFORMATIONS

Experiment 2.2

Verification of Even and Odd Signal Decomposition

Program :

% Even-Odd Decomposition of a Signal

```
clc;
```

```
clear;
```

```
close all;
```

```
% Time vector
```

```
t = -1:0.001:1;
```

```
% Original signal
```

```
x = exp(-abs(t)).*sin(4*pi*t) + 0.3;
```

```
% Reverse signal x(-t)
```

```
x_rev = interp1(t, x, -t, 'linear');
```

```
% Even and Odd parts
```

```
xe = 0.5 * (x + x_rev);
```

```
xo = 0.5 * (x - x_rev);
```

```
% Plot the signals
```

```
figure;
```

```
plot(t, x, 'k', 'LineWidth', 1.5);
```

hold on;

```
plot(t, xe, 'b--', 'LineWidth', 1.5);
```

```
plot(t, xo, 'r:', 'LineWidth', 1.5);
```

hold off;

grid on;

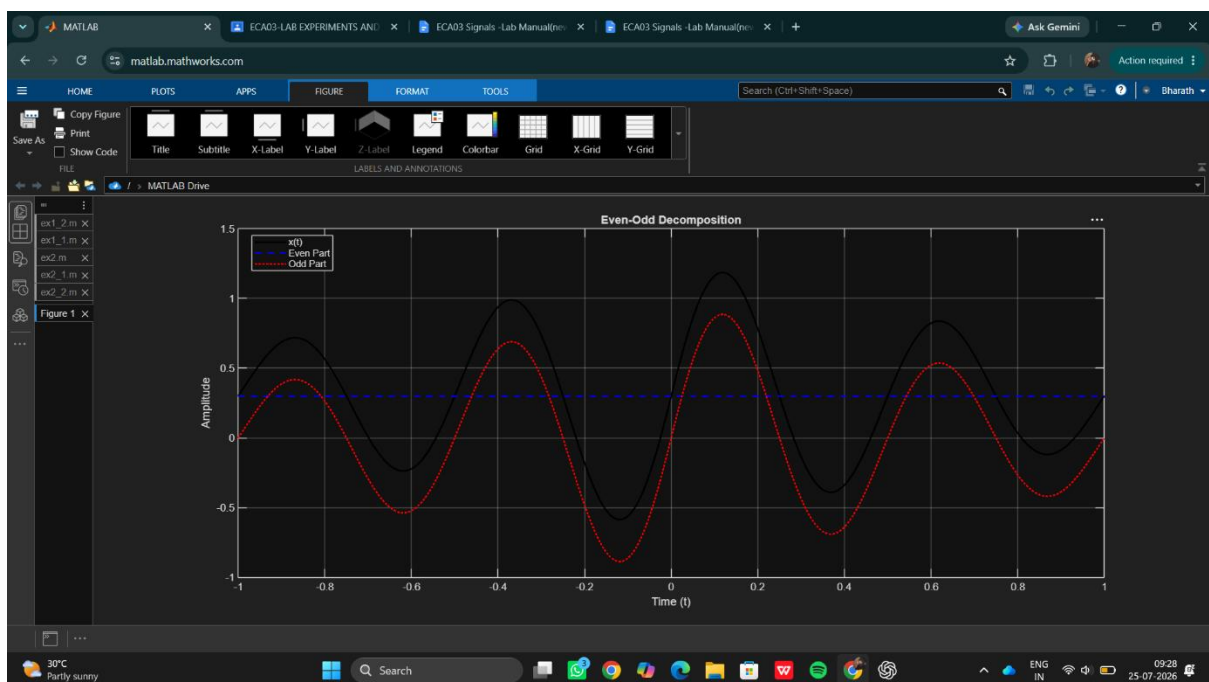
```
legend('x(t)', 'Even Part', 'Odd Part', 'Location', 'best');
```

```
title('Even-Odd Decomposition');
```

```
xlabel('Time (t)');
```

```
ylabel('Amplitude');
```

Output waveform :



Result

Signal equals sum of even and odd parts.